a. The cube code (9 min)

0–1.5 min / 1 Qubits indexed by the points of $\mathbb{F}_2^3$ Words are functions on the points. Steps: u1, u2, u3, then the constant word. They span RM(1,3).

1.5–3 min / 2 CSS codes from two nested binary codes Define the logical basis, the X and Z checks, and d_X, d_Z.

3–4.5 min / 3 The cube code $[\![8,3,2]\!]$ K has rows u1, u2, u3, which is why qubits are points. Z000 Z100 gives d_Z = 2.

4.5–5 min / 4 Transversal gates Product of single-qubit gates. A fault stays on one qubit. Define T and CCZ.

5–6 min / 5 The signed weight gives the phase T sites minus T-dagger sites. The gate condition mod 8.

6–7.5 min / 6 Choosing the signs on the cube p is the logical word of label 111. Distance to p is 4 for every other affine word.

7.5–8.5 min / 7 Weights of sums expand into overlaps The bit identity, summed over positions. Terms with four or more words vanish mod 8.

8.5–9 min / 8 The gate condition becomes conditions on generator rows Mod 8, mod 4, mod 2. Define the bracket. The cube passes.

b. History and the CH conditions (5 min)

9–11 min / 9 Historical context Cite the CSS-T work of Camps-Moreno, López, Matthews et al.

11–12.5 min / 10 Triorthogonal matrix and triorthogonal space Nezami-Haah puncture parent spaces. We fix S first and find K.

12.5–14 min / 11 Conditions for quasi-transversal CCZ Mod 2 reduction gives the nine parities. A physical layer needs the signed congruences.

c. Results and finite geometry (8 min)

14–15.5 min / 12 Result: a 48-qubit code with a transversal $T/T^\dagger$ CCZ 39 to 46 excluded by Wills, Jain and Singh.

15.5–17 min / 13 Fix S using finite/projective geometry code Four steps. Only the logical rows are searched.

17–18 min / 14 Projectivity rules out weight-one and weight-two Z logicals Parity-check argument. Covers d_Z only.

18–19.5 min / 15 Complement of a projective subspace 63 points, remove the 15 with w = 00, three blocks of 16.

19.5–21 min / 16 Hyperplane sections and codeword weights u1 gives 24, w1 gives 32. Both divisible by 8.

21–22 min / 17 The geometric construction produces a family p = 4, q = 2 gives 48. Lengths 48 to 496.

d. Column picture and search (9.5 min)

22–23.5 min / 18 Column indicators from the generator matrix Show G for the cube. chi_G = y because every x occurs with y = 1.

23.5–25.5 min / 19 The CH conditions are inner products with monomials Overlap parity is an inner product. Coset of RM(r-4, r).

25.5–27.5 min / 20 Logical rows in the column picture Define the logical label, T_K, T_S and the bracket. Step 2 shows K1.

27.5–30 min / 21 Finding the logical rows Algorithm 1: V_P, pairs, then Gaussian elimination for kappa 3. Witness gives d_Z = 3.

30–31.5 min / 22 Solving for the $T/T^\dagger$ pattern Binary unknowns, linear over Z4. 26 T plus 22 T-dagger.

e. Bounds and implications (8.5 min)

31.5–33.5 min / 23 Stabilizer supports are Reed–Muller codewords Indicator method from Nezami-Haah. Q48 gives a word of RM(2,6).

33.5–35.5 min / 24 Our exclusion argument covers every length through 38 Analytic reductions plus a complete finite enumeration.

35.5–37 min / 25 Concurrent classification closes the 39–46 gap CNOT plus S equivalence.

37–38.5 min / 26 What the 47-qubit comparison establishes No sign vector for the Jacinto code. Footprint numbers are theirs, not ours.

38.5–40 min / 27 Takeaways and open questions Leave the QR codes up during questions.

40 minutes: finish and invite questions.